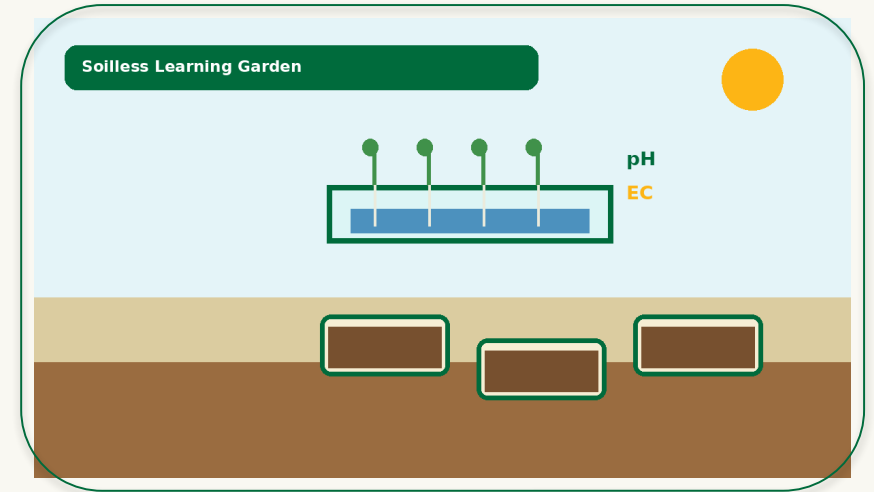


Soilless Learning Garden

Fundamentals of Hydroponic Production

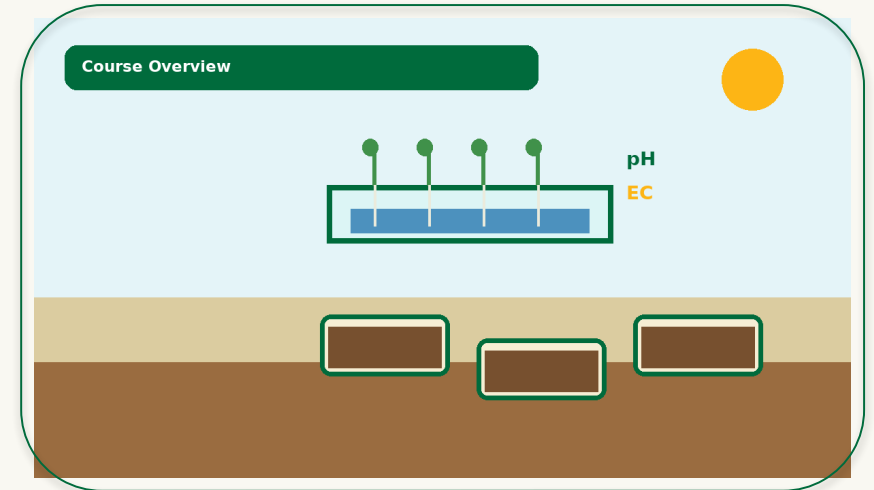
- LIFEWS AgriShine™ GrowMeal™ JSS 1-3 Teacher Training
- Audience: JSS 1-3 teachers, agricultural-science educators and garden facilitators
- Format: Self-paced online course + supervised practical/laboratory orientation
- Recommended duration: 3.5 hours online + 2 hours practical
- JSS learning progression: formulate -> test -> measure -> analyze -> manage



Course Overview

Learning outcomes, materials and teacher readiness

- Learning outcomes: explain the garden/laboratory system; teach all ten technical topics accurately; guide students in measurement, comparison and data collection; connect agricultural science with mathematics and systems reasoning; supervise practical work safely; maintain credible records; interpret evidence without overstating conclusions.
- Core materials: teacher guide, system/plot map, approved tools and irrigation equipment, labels, clipboards, rulers/tapes, graduated containers, meters/sensors where relevant, data sheets, clean sample/harvest containers, PPE/handwashing supplies and first-aid access.
- Completion expectation: complete all modules, demonstrate one safe practical session, and submit a lesson, measurement protocol, experiment or systems investigation appropriate to the course.



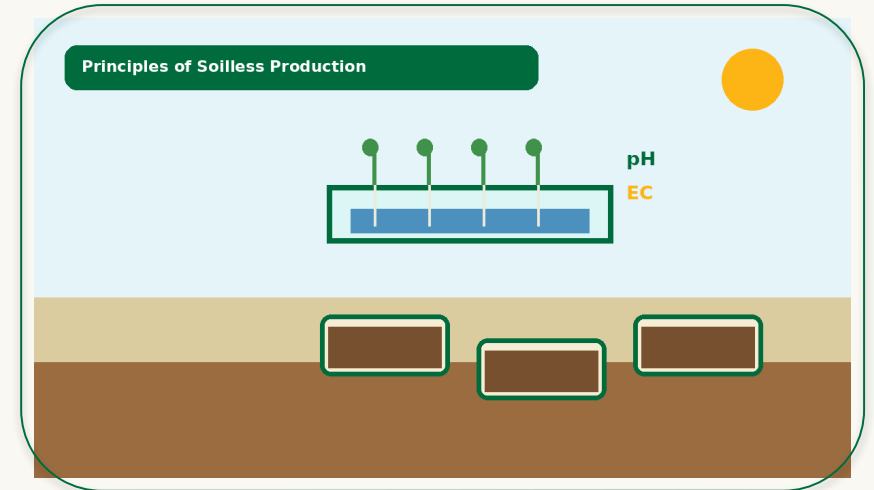
TEACHER PRACTICE

- Training sequence: QUESTION -> PLAN -> IMPLEMENT -> MEASURE -> ANALYZE -> DECIDE.

Principles of Soilless Production

Teacher knowledge, technical practice and student supervision

- Define soilless production as crop growth without field soil using managed root-zone water, nutrients and/or substrates.
- Identify system components: reservoir, root zone, crop support, water/nutrient delivery and drainage/return where applicable.
- Explain the need to manage water, oxygen, nutrients and sanitation more directly than in many soil systems.
- Compare open and recirculating concepts at an introductory level.
- Keep electrical and chemical components under authorized adult control.



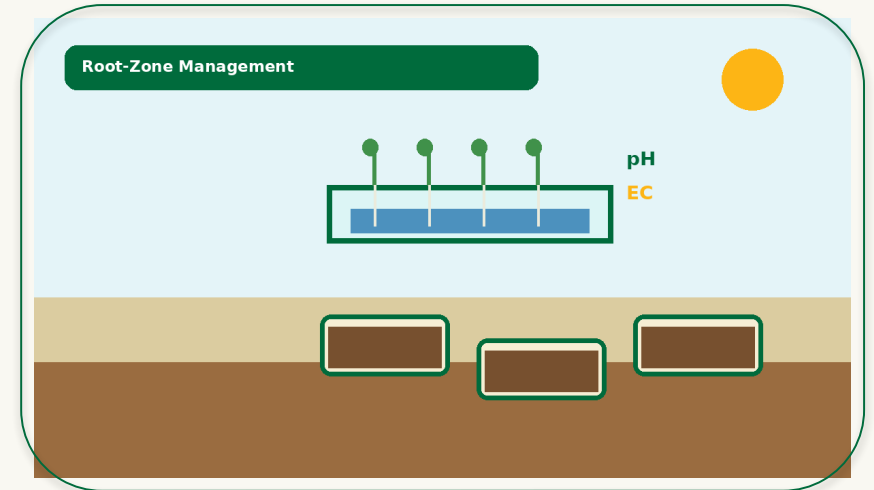
TEACHER PRACTICE

- Teacher task: trace inputs, root zone and outputs in the installed system.

Root-Zone Management

Teacher knowledge, technical practice and student supervision

- Explain the root zone as the environment where roots access water, oxygen and nutrients.
- Monitor root-zone moisture or solution level according to system type.
- Observe root color and condition through safe access points without repeated disturbance.
- Prevent prolonged waterlogging in substrate systems and inadequate solution contact in water systems.
- Record root-zone problems alongside environmental conditions.



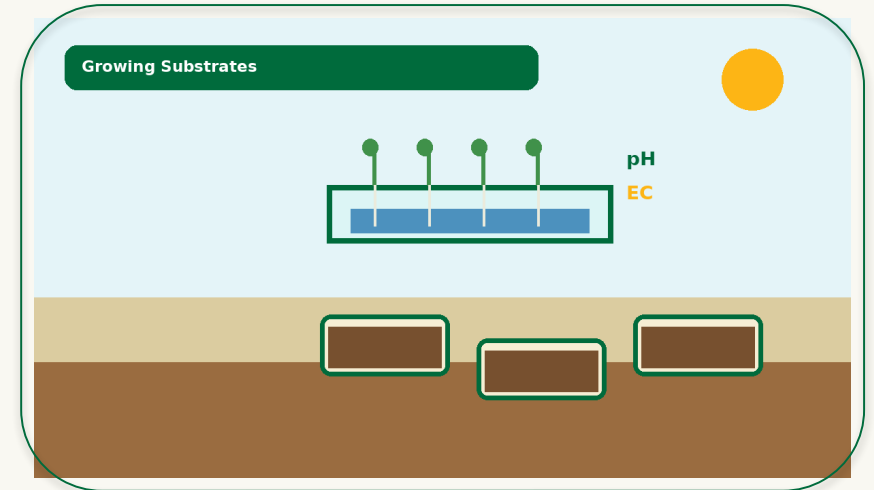
TEACHER PRACTICE

- Prepare a root-zone inspection checklist.

Growing Substrates

Teacher knowledge, technical practice and student supervision

- Compare approved substrates by support, porosity, water retention, drainage and handling characteristics.
- Examples may include coco coir, perlite or other locally available horticultural media used by the system.
- Explain that most inert substrates do not supply complete crop nutrition.
- Handle dusty materials using appropriate precautions.
- Avoid untested waste materials in food-production systems.



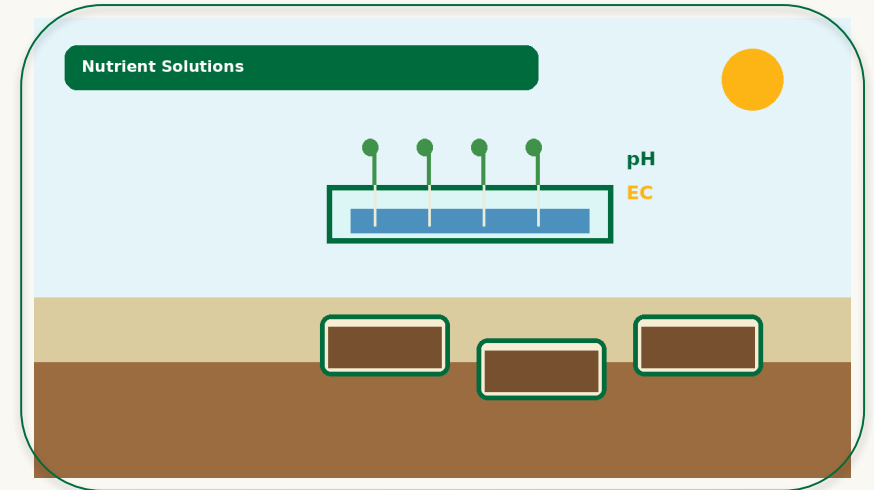
TEACHER PRACTICE

- Create a substrate comparison table.

Nutrient Solutions

Teacher knowledge, technical practice and student supervision

- Explain nutrient solution as water containing dissolved plant nutrients at managed concentrations.
- Authorized adults prepare concentrates and final solutions according to the system recipe.
- Introduce stock solution and dilution concepts without allowing pupils to handle concentrated chemicals.
- Record formulation, date and reservoir volume.
- Never assume more concentrated solution produces better growth.



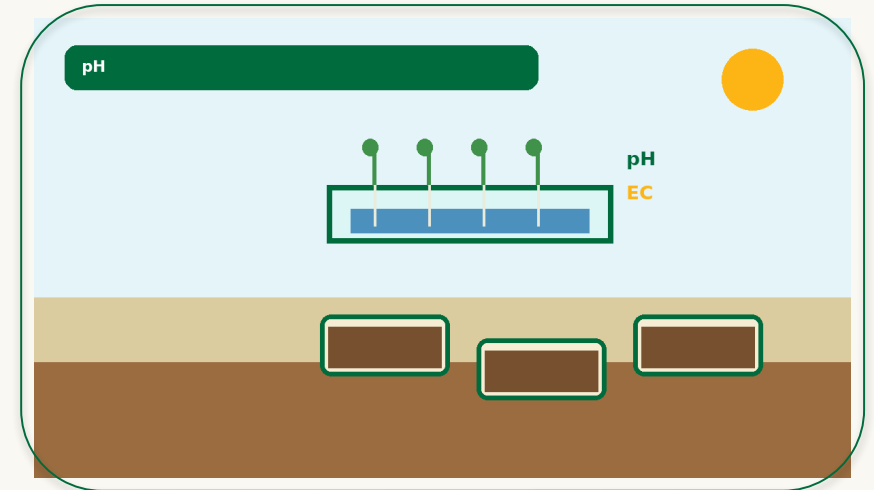
TEACHER PRACTICE

- Demonstrate the nutrient pathway using water or a safe model rather than concentrates.

pH

Teacher knowledge, technical practice and student supervision

- Define pH as a measure related to acidity/alkalinity that influences nutrient availability and system chemistry.
- Use calibrated meters or suitable test kits according to manufacturer instructions.
- Record pH with date, time and system identity.
- Only authorized adults add pH-adjustment chemicals.
- Explain that target ranges depend on crop and system guidance.



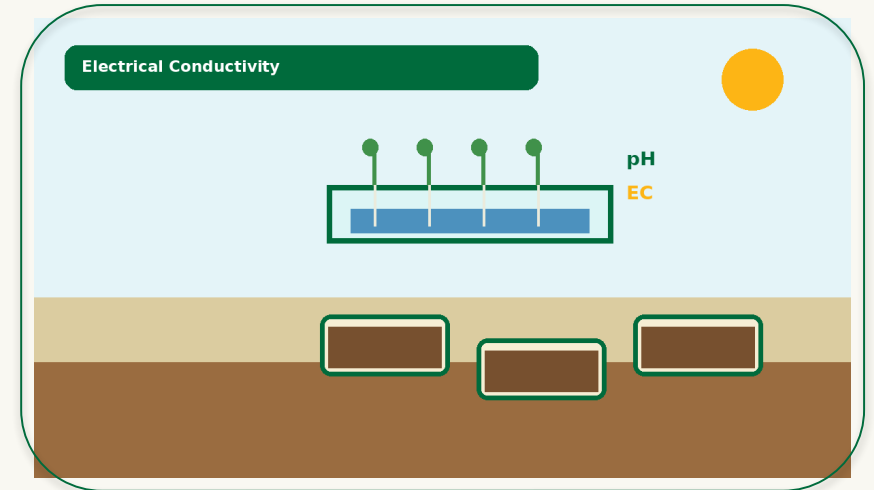
TEACHER PRACTICE

- Practice meter reading and recordkeeping using teacher-controlled samples.

Electrical Conductivity

Teacher knowledge, technical practice and student supervision

- Define electrical conductivity (EC) as a measure of a solution's ability to conduct electricity, commonly used as an indicator of total dissolved ionic concentration.
- Use a calibrated EC meter and record units exactly as displayed.
- Explain that EC does not identify individual nutrients.
- Interpret readings against the system's crop-specific management range.
- Authorized adults make nutrient adjustments.



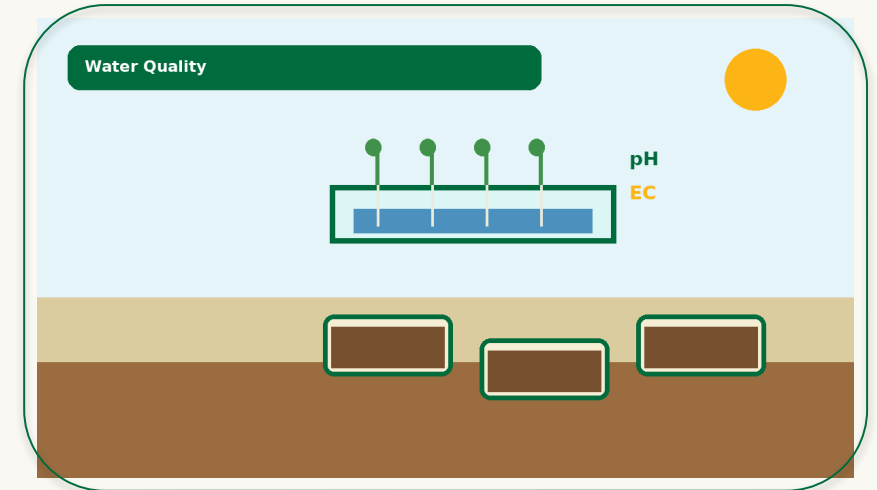
TEACHER PRACTICE

- Prepare an EC measurement and interpretation exercise.

Water Quality

Teacher knowledge, technical practice and student supervision

- Discuss source-water properties that can affect hydroponic management, including pH, EC/salinity and cleanliness.
- Use available water-quality information or simple measurements.
- Recognize that clear-looking water is not necessarily suitable or microbiologically safe.
- Keep potable and nutrient-solution equipment separate where required.
- Record source changes that may affect system performance.



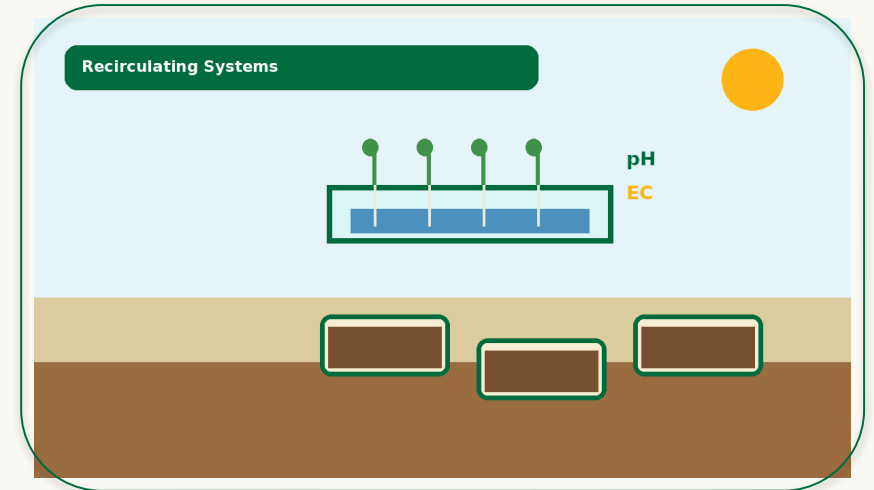
TEACHER PRACTICE

- Develop a basic source-water record.

Recirculating Systems

Teacher knowledge, technical practice and student supervision

- Trace solution from reservoir to crop root zones and back to the reservoir.
- Explain potential benefits of reuse alongside the need for monitoring and hygiene.
- Check pumps, return lines, leaks and reservoir levels under adult supervision.
- Recognize that problems can spread through a shared recirculating solution.
- Keep electrical equipment inaccessible to students.



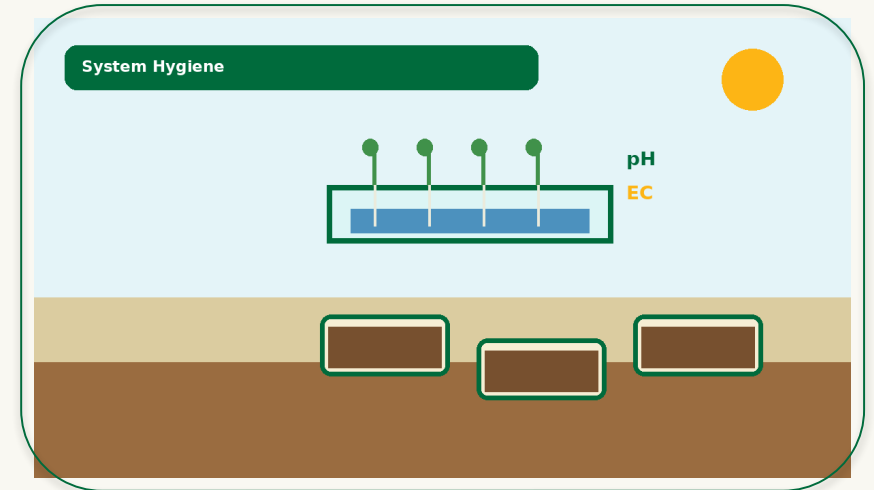
TEACHER PRACTICE

- Draw the recirculation loop and identify monitoring points.

System Hygiene

Teacher knowledge, technical practice and student supervision

- Clean reservoirs, channels, tools and contact surfaces according to a documented schedule.
- Remove dead plant material promptly and prevent algae/mosquito access.
- Avoid cross-contamination between healthy and suspect plants.
- Adults manage disinfectants and electrical isolation.
- Maintain handwashing and clean harvest procedures.



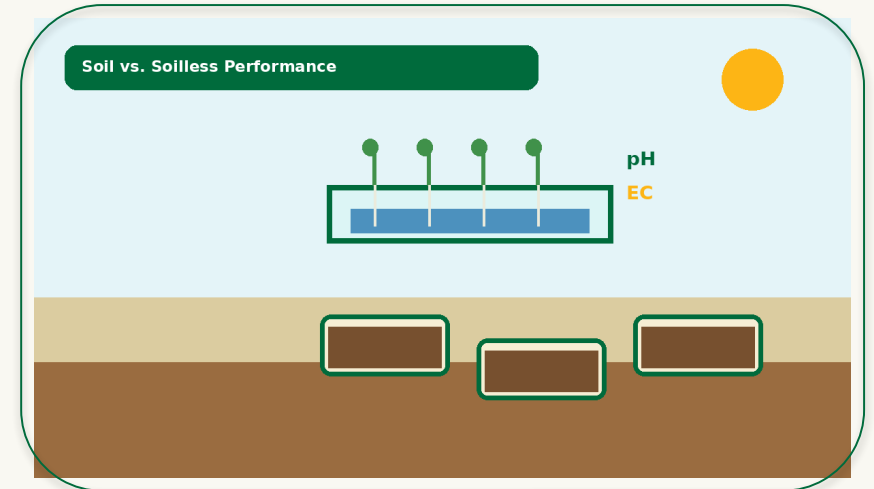
TEACHER PRACTICE

- Prepare a hydroponic hygiene SOP.

Soil vs. Soilless Performance

Teacher knowledge, technical practice and student supervision

- Grow the same crop in comparable soil and soilless treatments where feasible.
- Measure growth, yield, water input, labor and system-management requirements.
- Record differences in plant population and environment.
- Avoid declaring one system superior from a single indicator or short trial.
- Discuss performance in relation to the objectives and constraints of each system.



TEACHER PRACTICE

- Completion task: design a soil-versus-soilless performance investigation.